Project Description

1. Intelligent E-Commerce Recommendation System

Project Description

Design and develop an advanced intelligent recommender system for an e-commerce platform that generates personalized product recommendations using three different recommendation approaches.

The system should simulate how real-world platforms provide intelligent suggestions by combining multiple techniques and allowing comparison between them.

Unlike basic recommender systems, your solution must implement and apply all three approaches:

- Collaborative Filtering
- Content-Based Recommendation
- Knowledge-Based Recommendation

The system should allow users to explore how recommendations differ depending on the selected approach and analyze which methods perform better under different conditions.

The goal is to build a system that is realistic, flexible, explainable, and analytically evaluated.

System Requirements

1. Mandatory Multi-Approach Implementation

You must implement ALL THREE approaches:

- Collaborative Filtering
- Content-Based Recommendation
- Knowledge-Based Recommendation

Each approach must:

- Be functional
- Generate recommendations
- Produce different results for the same user

2. Multiple Methods for Comparison

You are required to implement at least 4 different methods within Collaborative Filtering (mandatory).

Additionally:

- You may implement multiple methods in Content-Based and/or Knowledge-Based (optional but recommended).

These methods should reflect different ways of computing similarity, ranking, or decision-making, enabling meaningful comparison.

3. Evaluation & Comparison

The system must include a comprehensive evaluation and comparison module.

You are required to:

- Use at least 4 different evaluation techniques
- Compare:
  - Between Collaborative Filtering methods
  - Between the three approaches (CF vs Content-Based vs Knowledge-Based)

You must clearly answer:

- Which method performs best?
- Which approach performs best?
- Under what conditions does each perform better?
- Why do differences occur?

The focus is on analysis and reasoning, not just displaying results.

4. Explanation & Interpretability

The system must provide clear explanations for recommendations, such as:

- "Recommended because similar users purchased this item" (CF)
- "Recommended because it matches your preferred category" (Content-Based)
- "Recommended based on your selected requirements (budget, brand, etc.)" (Knowledge-Based)

5. Expected System Behavior

The system should:

- Accept user inputs (ratings, purchase history, preferences, constraints)
- Allow selection of recommendation approach
- Generate recommendations using:
  - CF
  - Content-Based
  - Knowledge-Based
- Display results clearly
- Provide explanations
- Show evaluation and comparison results

Project Notes

- The project must be completed in teams of up to 5 students.
- The final discussion and presentation will take place in Week 13.
- All team members must be aware of all project phases, not only their individual parts.
- Any form of copying or plagiarism will result in penalties according to course regulations.